

artistic style; it *lacks* a style. Its photorealism, foreshortening, negativity, and absence of outline and light focus are unknown in pre-Impressionist art. A skeptic need only look at one of the more than forty extant copies of the Shroud to realize how poorly the image was understood by the best artists of the day (figures 4 and 19). Yet Craig and Bresee, the researchers who proposed the carbon-dusting technique, amazingly cite the existence of such copies as evidence that the Shroud too is perhaps a copy of Jesus' original burial shroud.²⁵² This bewildering assertion obviously misses the fact that these other "shrouds" are documented copies of the Turin Shroud; it also implies that the true shroud had an image on it, for why else would the copies have one?

Equally untenable (though at least acceptant of the fact that the Shroud enfolded a corpse) is the assertion that the man in the Shroud was a fourteenth century crucifixion victim, perhaps a Jew ritually slain for the relic-production cause.²⁵³ Not only would the perpetrator have had to effectively practice crucifixion in his spare time in order to be aware of all the subtleties of the execution, but he would have had to acquire, as his victim, a subject whose facial features somehow incorporated those of hundreds of icons. He would still have had to obtain a pollinated burial shroud and buried the victim according to first-century Jewish practices. Most significantly, he would have had to invoke some unknown image-formation process. It does not take a coroner to know that dead bodies don't produce images on cloth, and actively modified (oiled, powdered, heated, etc.) corpses don't leave images remotely resembling that of the Shroud. But our executioner would nonetheless have succeeded in creating a relic whose very existence (the image, that is) receives nary a mention in the gospel accounts; furthermore, as the late biblical scholar John A. T. Robinson noted, the harmony between the Shroud and the gospels is so simultaneously precise and non-intuitive that "no forger starting, as he inevitably would, from the details of the Gospels, and especially that of the fourth, would have created the shroud we have."²⁵⁴

Such are the logical consequences of upholding the radiocarbon verdict, yet accusations of "will to believe" and "blind to the facts" are more typically hurled at those who maintain that the Shroud is neither medieval nor artistic. Ted Hall of the Oxford Radiocarbon Laboratory has repeatedly declared that anyone who favors the Shroud's authenticity should join the Flat Earth Society. But is a skepticism about the anomalous

radiocarbon dating really more ignorant than Hall's claim that "someone just got a bit of linen, faked it up, and flogged it?"²⁵⁵

The *only* data supportive of a medieval origin are the d'Arcis memorandum and the carbon dating. The historical reliability of the first has been seriously questioned, and the latter stands in conflict with a confluence of evidence at least as scientifically significant. As one researcher put it, "do you really believe we must start all over again from the beginning?"²⁵⁶

The testimony of pathology, microchemistry, and image analysis is that the Shroud is a shroud; i.e., it enfolded a bleeding, crucified body. Ian Wilson's historical reconstruction has received support in the work of a number of other researchers.²⁵⁷ In conjunction with iconographic studies, these provide for a relatively firm historical judgment that the Shroud existed as early as the sixth century C.E. Roman crucifixion was not practiced after the third century, and the presence of an amulet and two coins dating from the reign of Tiberius Caesar effectively date the crucifixion of the man in the Shroud to no later than the second half of the first century C.E.²⁵⁸ The preponderance of data collected over several decades of research on the Turin Shroud warrants the claim that the cloth should, in the words of archaeologist William Meacham, be considered "an archaeological document of crucifixion."²⁵⁹ The question of the identity of the man in the Shroud is necessarily less scientifically absolute; there is, to paraphrase researcher Al Adler, no "test for Christness."²⁶⁰ This is not to say, however, that this question of identity is inherently unresolvable.

The Identity of the Man in the Shroud

When the Shroud is assumed to be the work of an artist, no one disputes the identification of its subject as Jesus of Nazareth; yet when it has been established that the Shroud is an authentic burial cloth, the matter of identification becomes more complicated. This is curious, but not unduly so: just because there is only one individual recorded to have suffered in the manner the gospels describe (and thus famous enough to have his image painted,) it does not mean there absolutely were no others. There are strong historical-theoretical objections, however, to the suggestion that the man in the Shroud was someone other than Jesus.

The physiognomy of the man in the Shroud suggests he was Semitic, and the details of the burial (use of long linen sheet, jaw-band, closing of eyes, binding of hands, general burial posture, body actually unwashed) point to a Jew buried by other Jews. The stigmata (marks from *flagrum*, *lancea*)

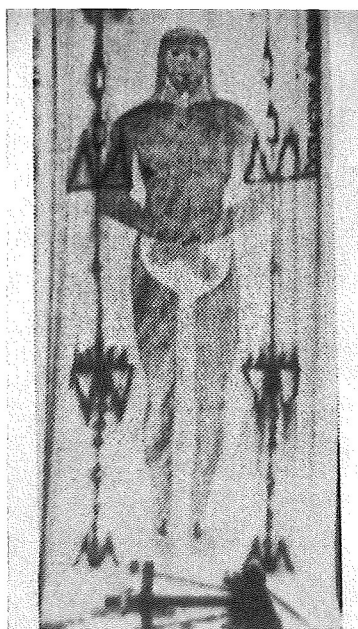


Figure 19. Detail of a 1697 copy of the Shroud, now in a Carmelite Monastery in Savona, by Giovanni Battista Fantino. At bottom the artist has drawn some symbols of the Passion, not appearing on the Shroud. This copy of the Turin cloth is typical of medieval copies of the Shroud image in its primitive attempt at reproducing the relic's negativity and fine shading. The addition of a "modesty cloth" and crude treatment of the wounds (for some unknown reason, the wound in the side appears on the left in this copy, even though other copies by the same artist render the wound in its correct position on the right) are also typical.

indicate a specifically Roman crucifixion. The individual burial (as opposed to the deposition in a mass grave) and quality of the linen cloth are unusual for a crucifixion victim, and suggest that he was neither a slave nor a common criminal. The presence of rigor mortis and lack of decomposition indicate that the man was removed from the cloth between one and three days after death.

The most obvious association of the man in the Shroud with Jesus of Nazareth is the correspondence between the inferred sufferings of the former and the sufferings of Jesus as recorded in the gospels. The crowning with thorns is unique in all recorded history, and there is only one other reference to a possible side-piercing during a crucifixion.²⁶¹ Several scholars have "calculated" infinitesimal probabilities for the possibility that someone other than Jesus could have been crucified and buried in the exact same way as he, but such calculations mean little since, if the Shroud is not Jesus', we are most likely not dealing with a set of independent events; i.e., if not the Nazarene's, the Shroud is most likely the burial cloth of someone who was deliberately executed in the same manner as Jesus, perhaps out religious fanaticism.

As reasonable as this sounds, several medical and historical facts make it a near impossibility. While the blood flows on the scalp correspond to puncture wounds inflicted on a living man, the flow from the side wound is clearly a postmortem issue of blood from an upright corpse. The puncture wounds on the scalp could not have been formed by mutilating a corpse, nor could the wound in the side have been inflicted on a living individual. The positioning of the side wound necessary to force the issue of "blood and water" as described in the gospels and seen on the Shroud had to be either extremely precise or fortuitous. Even deliberate attempts to mimic the crucifixion and sufferings of Jesus would likely fail on one or more points. Furthermore, such an execution in the early centuries C.E. would presumably have been carried out in order to produce a "true" burial *sindon* of Jesus, but the demand for relics did not become significant until after Constantine outlawed crucifixion.²⁶²

An extra-biblical connection between the man in the Shroud and Jesus of Nazareth has been provided by studies of the depictions of Jesus in the Roman catacombs. Australian researcher Rex Morgan has found that a few of these paintings, some of which date to the latter part of the first century C.E., strongly resemble the man in the Shroud. It is believed that these paintings were made by those with first- or second-hand knowledge of Jesus' appearance, in effect providing the only known physical description of the Nazarene.²⁶³ The apparent conclusion is that if the man in the Shroud was not Jesus, it was someone who looked quite a bit like him.

Perhaps the strongest argument against the identification

of the Shroud with a crucifixion victim other than Jesus is the relic's very existence. The body of the man in the Shroud was removed from the cloth within a few days after death. The problem is the same as that suffered by suggestions of a fourteenth century crucifixion: there is no apparent reason for the body to have been thus removed. Why would anyone, whether relic-monger or zealot, anticipate an image when no other such image has ever been formed by a corpse?

The identification of the man in the Shroud as Jewish makes the situation even more troublesome. It is almost inconceivable that Jews, in light of the ritual uncleanness of a corpse, would remove the body from the cloth. It is just slightly less inconceivable that, if the burial cloth were discovered empty, it would be preserved. The objective facts of the early removal of the corpse and the preservation of the cloth force one to posit a highly unusual post-enshroudment scenario. Of course, the sequence of events related in the gospels is rather unusual.

"A RELIGIOUS QUESTION HAS BEEN NEEDLESSLY INJECTED INTO A PROBLEM WHICH IN ITSELF IS PURELY SCIENTIFIC, WITH THE RESULT THAT FEELINGS HAVE RUN HIGH, AND REASON HAS BEEN LED ASTRAY. IF, INSTEAD OF CHRIST, THERE WERE A QUESTION OF SOME PERSON LIKE A SARGON, AN ACHILLES OR ONE OF THE PHARAOHS, NO ONE WOULD HAVE THOUGHT OF MAKING ANY OBJECTION...."

Regardless of how the body of Jesus disappeared, the story of the empty tomb, the discovery of the burial linens, and the subsequent reactions of the disciples provides a framework in which both the preservation of the cloth and the silence on the matter of its image can be reasonably accounted for; the preservation of the Turin Shroud can be explained if it is the burial cloth of Jesus and the events de-

scribed in the gospels (resurrection aside) actually occurred. It is exceedingly difficult to explain the existence of the Shroud, and the initial silence about its image, in another context.

Just as it is unreasonable to claim that the radiocarbon dating provides proof of the Shroud's fraudulence, it is unreasonable to claim that these facts provide incontrovertible scientific proof for the identification of the man in the Shroud as Jesus; no responsible investigator has done so. It is manifestly not the case, however, that the question of identity is a hopeless puzzle surrounded by much conjecture and little certainty. Historical judgments, conclusions, and identifications are consistently made on the basis of evidence not half as substantial as that in favor of the Shroud's authenticity. Many researchers as well as lay followers of the ongoing Shroud saga are often quick to point out that despite the lack of any viable candidates besides Jesus of Nazareth, the identity of the man in the Shroud can never be proven. In a purely semantic sense this is true, barring of course any heretofore unknown "test for Christness." But as archaeologist Meacham explains, historical conclusions are formed within an inductive framework distinct from purely scientific methodology:

"The identification and dating of artifacts by their cultural affinities is part and parcel of archaeology. A web of intricate, interlocking, field-tested evidence is usually

taken as proof, though not exactly comparable to proof in the natural sciences...."²⁶⁴

The lack of absolute certainty in historical proclamations is an epistemological consequence of the irreversible passage of time; this concession of the absolute is so fundamental to historical and critical reasoning that it is typically taken for granted until its very recognition serves the argument at hand.

The hesitancy among even sympathetic researchers to proclaim the Turin Shroud authentic no doubt arises at least in part from the emotional reactions such a proclamation would incite. The identification of the Shroud as the burial cloth of the man central to a faith which claims nearly one-third of the world's population as adherents would, in all likelihood, interest more than a few armchair enthusiasts. But should the inevitable religious sympathies aroused by this identification necessitate the almost compulsive conservatism so prevalent in pro-Shroud judgments? It is instructive to consider the words of the agnostic Yves Delage, written 90 years ago when his conclusions regarding the Shroud led to fabricated charges of scientific misconduct:

"A religious question has been needlessly injected into a problem which in itself is purely scientific, with the result that feelings have run high, and reason has been led astray. If, instead of Christ, there were a question of some person like a Sargon, an Achilles or one of the Pharaohs, no one would have thought of making any objection.... I have been faithful to the true scientific spirit, treating this argument with the sole intention to find the truth.... I consider the Christ to be an historical personage and I do not see why anyone should be scandalised that there exists a material trace of his existence."²⁶⁵

The philosophical climate of the 1902 French Academy of Sciences was hostile towards anything that reeked of religion. Contemporary scientific culture is perhaps even more so. This is precisely why Delage's point remains as relevant as when first stated: the authenticity of the Turin Shroud is a matter of scientific and historical judgment, and *not* of religious interpretation.

Yet while the authenticity of the Shroud should not stand or fall on its potential religious significance, it is naive to expect the public (including prospective sensationalists) to refrain from extracting such significance from the Shroud. Delage's insistence on the irrelevance of the religious question might not have been so emphatic if the vaporograph theory, which he and Vignon embraced as the cause of the cloth's image, had been as thoroughly discredited then as it is today.²⁶⁶ At the heart of the sindonological puzzle stands the fact that modern science has been unable to provide an adequate explanation for the cloth's unique image. This uniqueness—echoed in the pronouncement by some of the primary researchers that "to our knowledge no other image on any cloth—grave cloth or art form—like the body image on the Shroud is known to exist today."²⁶⁷—is due not to mere lack of reproduction, but rather to apparent impossibility of reproduction. The authors of one of the more comprehensive reviews of the Shroud image write: "The dilemma is not one of choosing from among a variety of likely transfer mechanisms but rather that no technologically-credible process has been postulated that satisfies all the characteristics of the existing image."²⁶⁸ An even greater

dilemma is that, taken together, the independently verifiable characteristics of the Shroud image seem to preclude any process operative within our current scientific understanding of the physical universe.

Were the Shroud identified as the burial cloth of Jesus, then, coupled with the fact of the removal of the body between two and three days after death, the lack of a viable image formation mechanism would be elevated from a scientific curiosity to a potential thorn in the side of secular authority. Can the Shroud offer "proof" of the resurrection? This often loaded question is ultimately meaningless, for an affirmative response implies knowledge of the processes involved in a "resurrection." Were the Shroud identified as the Nazarene's, one can still only go so far as to say that it corroborates the gospel story of the removal of the body from the grave cloths and testifies to a possibly unique set of circumstances surrounding this removal.

If the nature of the image on the Shroud is at the heart of the linen enigma, the relationship between science and the Shroud constitutes the most intriguing sindonological subplot. Before the photography of the cloth in 1898 the "Holy" Shroud remained a religious relic of little interest to any but fetishistic devotees. In the century since, it has become an object of extensive scientific and historical inquiry—potentially the most intensely studied single object in all of human history. The real curiosity in this still-evolving drama has not been the novelty of science investigating an object of potential religious significance, but rather the battery of new questions raised by each successive wave of technological analysis. Decades of scientific investigation have supplied answers to such problems as the composition of the bloodstains and whether the cloth enfolded a human body; these foundational issues have been so rigorously investigated that according to two commentators, "in the twentieth century, some scientists accept the Shroud's authenticity more readily than medieval Christians did."²⁶⁹ Yet the most pressing Shroud-related riddle remains just that: the more data that is collected, the more puzzling the image becomes. This twist in the paradigm of modern scientific investigation is as unique as the image itself.

Scientific investigation has shown that the Shroud image is, like the sacred face of the Edessan Mandylion, *acheiropoiotos* - not made by hands. This fact continues to be met with a range of reactions, from reflexive dismissal to cautious acceptance to overenthusiastic extrapolation; the Shroud's status as the most intensely studied artifact in the history of modern science might not be as secure as a claim to being the most intensely debated artifact in the history of modern discourse. Yet few who look upon a photograph of the relic do not give pause at the prospect that, locked away in a Turin cathedral, lies a linen sheet on which a haunting image seems to echo with the charge: "Behold the man."

Notes

¹Meacham 1983:283 remarks that studies of the Shroud comprise "what must be the most intensive and varied scrutiny by scientific

- means of any archaeological or art object in history." Some estimate that as many as 250,000 man-hours have been spent scientifically studying the Shroud.
- ²The Shroud was moved from the Chapel of St. John to the Cathedral on 24 February 1993, to protect it from repairs being conducted on the Chapel structure. See Dorothy Crispino's "A Providential Disaster," *Shroud Spectrum International* No.42 (Dec. 1993):40-42.
- ³Schwalbe and Rogers 1982:6
- ⁴Matthew 27:59, Mark 15:46, Luke 23:53, John 19:40
- ⁵The complete text of Herbert Thurston's translation of the d'Arcis memorandum is reprinted in Wilson 1979:266-272. Fossati (1983 & 1992) has shown that Thurston's translation in regard to the "confession" of the artist is not consistent with the original text. There is no mention of a confession, and contrary to the most reports, the document is best translated as saying an artist had depicted (reproduced) the Shroud, rather than painted it.
- ⁶Of particular importance is the obvious slighting d'Arcis was subject to in being bypassed by de Charny who went directly to the pope for permission to display the relic (Wilson 1979:86, Stevenson and Habermas 1990:154-156). Of supreme importance is the fact that d'Arcis neither names the hypothetical artist nor cites a documented confession, both of which would be expected in an official appeal to the pope. Furthermore, the "memorandum" is in reality a rough draft, and there are no records of it ever having been sent to the papacy (Fossati 1992).
- ⁷This holland cloth was not separated from the Shroud until 1978, when it was temporarily pulled away so the back of the relic could be examined. It remains attached to the Shroud. See Dorothy Crispino 1982, Jumper et. al. 1984:466.
- ⁸This overview of the modern history of the Shroud is taken primarily from Wilson 1979:84-92, 191-223, 259-265.
- ⁹For a detailed discussion of Pia's photography of the Shroud, see Wilson 1979:26-28.
- ¹⁰See Van Cauwenberghe 1992:13-19 and Wilson 1979:32-34 for accounts of the events surrounding the collaboration of Vignon and Delage and also of Delage's presentation to the Academy and its aftermath.
- ¹¹Carleton Coon in Stevenson and Habermas 1981:35, also T. Dale Stewart in same.
- ¹²Daniel-Rops 1962:299 and Gressman as cited by Wilson 1979:47 identify the pigtail as one of the most common fashions for Jewish men in antiquity. The identification of the pigtail on the Shroud image has been questioned (Pellicori in Meacham 1983:301, "Comments," attributes it to "nonuniform image-resulting chemistry as the weave direction changes") but evidence from the VP-8 analysis of the Shroud (see section on STURP investigations) suggests that there is a definite three-dimensional structure to the object.
- ¹³Many estimates of the man's height, ranging from about 5'7" to 6'1" have been made, but a man 5'10" tall would best fit the Shroud's image (Ercoline et. al. 1982). The 1968 excavation of a first century Jewish cemetery in East Jerusalem has yielded information about the average height at the time, though the sampling is unfortunately rather small. The average height of 9 mature (over 20 years) male skeletons was just over 5'6", but one male was between 5'8"-5'9" and another was over 5'11" (Haas 1970).
- ¹⁴Stevenson and Habermas 1981:59-60
- ¹⁵Stevenson 1977 contains the text of the papers presented at this conference.
- ¹⁶Stevenson and Habermas 1981:65-66
- ¹⁷Damon et. al. 1989:611
- ¹⁸Raes 1991:6
- ¹⁹Wilson 1979:69
- ²⁰This is consistent with the gospel accounts of the burial of Jesus, which state that Joseph of Arimathea, a rich disciple and member of the Sanhedrin, purchased the linen cloth in which the body of Jesus was wrapped. Matthew 27:57-59, Mark 15:42-46, Luke 23:50-53, John 19:38-40.
- ²¹Schwalbe and Rogers 1982:42-43, Crispino 1990. The use of the same loom to weave both cotton and linen would not violate the Levitical prohibition of mixing of kinds, and would not therefore render the Shroud unacceptable for Jewish use. (Leviticus 19:19, Deuteronomy 22:11)
- ²²Wilson 1979:77-78
- ²³Pollen grains are both resilient and distinctive; their hard outer shells allow individual grains to survive millions of years, and the characteristics of those shells allow precise floral identifications.
- ²⁴Frei actually identifies 16 species of plants from Anatolia, but ten of these are more abundant in Israel and are counted among the desert plants (Frei 1982:6-7).
- ²⁵As quoted in Meacham 1983:288; pollen distribution statistics from Frei 1982:5-7.
- ²⁶Frei and researcher Werner Bulst as quoted in Stevenson and Habermas 1990:63-65. Also, Frei 1982 and Bulst 1988. Wilson 1979:297-298 notes an experiment which determined that under typical conditions, 95 percent of pollen gathered by a viscous plate placed in the open air originates from plants growing within an area of a few hundred meters. Compare this to the approximately 2500 kilometers from France to Israel. Even more significant, Bulst 1988:14 notes that there are no Mediterranean wind patterns that correspond to the path the pollens would have to have taken to travel from Israel to France or northern Italy.
- ²⁷Frei 1982:7
- ²⁸Personal communication with Paul Maloney, Director, Ancient Near Eastern Researches (Quakertown, Pennsylvania).
- ²⁹Jackson et. al. 1977:74-77. This correlation between cloth-body distance and image intensity also accounts for the negativity of the Shroud image.
- ³⁰Jackson et. al. 1977:85; Jackson, Jumper, and Ercoline 1984
- ³¹Jackson 1988:5
- ³²Lorre and Lynn 1977:177
- ³³Tamburelli 1981:670-676, for example
- ³⁴Jumper and Mottern 1980:1911
- ³⁵Accetta and Baumgart 1980, Gilbert and Gilbert 1980, Pellicori 1980
- ³⁶Morris, Schwalbe, and London 1980
- ³⁷Mottern, London, and Morris 1980
- ³⁸Accetta and Baumgart 1980
- ³⁹Miller and Pellicori 1981
- ⁴⁰Pellicori and Evans 1981
- ⁴¹Heller and Adler 1980, Heller and Adler 1981. The samples were taken from the surface of the cloth by the application of Mylar tape with a special polymer adhesive. 36 tapes, approximately 5 square cm each, were taken from pre-identified locations on the Shroud, providing a range of data for both image and non-image areas.
- ⁴²Several comprehensive reviews of the physics and chemistry of the Shroud have been made. The reader is referred to the very-readable Jumper et. al. 1984, and to Schwalbe and Rogers 1981. Dinegar 1982 is a more concise presentation of the broader results.
- ⁴³The Shroud is thus analogous to a halftone; there are only two degrees of coloration of the fibrils of the cloth: yellowed and not yellowed.
- ⁴⁴The threads of the Shroud are typically at least .15 mm in diameter, and are made up of linen fibrils 10-15 mm in diameter. The body-only image penetrates only one or two fibrils deep. Schwalbe and Rogers 1982:11, Jumper et. al. 1984:450, Meacham 1983:288.
- ⁴⁵Jumper et. al. 1984:456-458
- ⁴⁶*ibid.*, pp.458-459.
- ⁴⁷*ibid.*, p.459; author's conversations with Al Adler.
- ⁴⁸Miller and Pellicori 1981:85, Jumper et. al. 1984:460, Heller and Adler 1981:96, Dinegar 1982:10
- ⁴⁹Heller and Adler 1981:91. Thus the proteinaceous coatings protected the underlying fibers from whatever process it was that induced the chemical reactions responsible for the image on the cloth. Jumper et. al. 1984:460.
- ⁵⁰Heller and Adler 1981:92-99
- ⁵¹Pellicori 1980; Gilbert and Gilbert 1980; Morris, Schwalbe, and London 1980; Miller and Pellicori 1981.
- ⁵²Immunochemical results reported in Bollone et. al. 1983:3-5. Potential cross-serological interactions noted in Jumper et. al. 1984:462. Problems with blood-type determinations discussed in Stevenson and Habermas 1990:158-159.
- ⁵³"Preliminary results of the 1978 Shroud of Turin investigation," Brochure from the Visual Data Exhibition at the University of Connecticut (11 October - 9 December 1981), p.2.
- ⁵⁴Jumper et. al. 1984:448
- ⁵⁵McCrone 1981
- ⁵⁶In Meacham 1983:298, "Comments."
- ⁵⁷Morris, Schwalbe, and London 1980
- ⁵⁸The retting process involves immersing the flax in water for an extended period of time, during which fermentation and natural ion exchange occur. The ions which bind most strongly to the flax fibers are Ca and Fe, and it is precisely these two elements that are found in the greatest concentrations on the Shroud. Heller and Adler 1981:97-98; Morris, Schwalbe, and London 1980:40-47. The charring of blood into iron oxide was first observed by Boyle in 1684, and explains the presence of iron oxide in the scorch areas. Heller and Adler 1981:97.
- ⁵⁹Jumper et. al. 1984:467-468, Pellicori and Evans 1981:42
- ⁶⁰*ibid.*, p.467
- ⁶¹Fossati 1984:36

- ⁶²Schwalbe and Rogers 1982:16
- ⁶³Morris et. al. 1980, Schwalbe and Rogers 1982:16-17
- ⁶⁴Gilbert and Gilbert 1980, Schwalbe and Rogers 1982:20-22
- ⁶⁵Miller and Pellicori 1981:84. Frei 1982:5 also noted that "none of the pollens was glued to the cloth with tempera or covered with tempera."
- ⁶⁶Schwalbe and Rogers 1982:23-4
- ⁶⁷Barbet's *A Doctor at Calvary* is quite readable, and though written nearly half a century ago, his conclusions are still regarded as accurate by many if not most sindonologists. Frederick T. Zugibe's *The Cross and the Shroud: A Medical Examiner Investigates the Crucifixion* (1982, New York: Angelus Books) differs at times with Barbet and is valuable as a comparative study. For a more concise study the reader might consult Bucklin 1982.
- ⁶⁸Because the left hand is folded over the right, only the puncture wound in the left wrist is visible. The blood flows seen on the right forearm, however, indicate a similar wound in the obscured portion of the right wrist.
- ⁶⁹John 20:25—all Scriptural quotes from the Revised Standard Version.
- ⁷⁰Stevenson and Habermas 1990:152
- ⁷¹Barbet carried out the most famous of these demonstrations. See Barbet 1963:106-120, Barbet 1987b:8-9.
- ⁷²Barbet 1963:116-118
- ⁷³Edwards et. al. 1986:1460
- ⁷⁴Zugibe, who also proposes a slightly different manner of placement of the nails, suggests that the position of thumbs folded into the palms is a natural one in death. (In Rhein, R. W., Jr. 1980. *The Shroud of Turin. Medical World News*, 22 December, pp. 40-50.)
- ⁷⁵Bucklin 1982:37, Barbet 1963:113-114, Edwards et. al. 1986:1461, Meacham 1983:285
- ⁷⁶Bucklin 1982:37, Barbet 1963:121-128
- ⁷⁷There is some disagreement in the literature regarding the number of contusions in each grouping.
- ⁷⁸Wilson 1979:48. An example of a *flagrum* was discovered during the excavation of Herculaneum, the Roman city buried with Pompeii in the 79 C.E. eruption of Mount Vesuvius.
- ⁷⁹Barbet 1963:45-46, Edwards et. al. 1986:1457
- ⁸⁰Matthew 27:26, John 19:1
- ⁸¹Wilson 1979:47-48, Meacham 1983:290,308
- ⁸²Stevenson and Habermas 1981:36, Meacham 1980:288
- ⁸³Tamburelli 1981, work of Lorre and Lynn as described in Wilson 1979:237
- ⁸⁴Miller and Pellicori 1981:75 and throughout.
- ⁸⁵As quoted in Wilson 1979:38-39
- ⁸⁶Bucklin 1982:36, Barbet 1987b:6-7, Judica-Cordiglia as cited in same.
- ⁸⁷Barbet 1963:37-71 makes a very thorough study of the historical practice of crucifixion, and most of his details are echoed by Edwards et. al. 1986:1458-1460. Barbet in particular discusses the meaning of "bearing the cross" as referring explicitly to the *patibulum*.
- ⁸⁸John 19:17
- ⁸⁹Matthew 27:32, Mark 15:21, Luke 23:26
- ⁹⁰The right knee is more severely wounded, suggesting it received the more weight in the fall. This is consistent with Barbet's (1963:99-101) argument that the difference in height of the left and right shoulder abrasions is illustrative of the natural position for the crossbeam to take when carried by a right-handed individual.
- ⁹¹Matthew 27:29, Mark 15:17, John 19:2
- ⁹²Meacham 1983:287-88
- ⁹³Barbet 1987b:6
- ⁹⁴See Sebastiano Rodante's "The Coronation of Thorns in the Light of the Shroud," *Shroud Spectrum International* No.40 (December 1991):14-15.
- ⁹⁵Barbet 1987b:4, Bucklin 1982:38, Lewis 1979:21
- ⁹⁶Wilson 1979:36
- ⁹⁷Bucklin 1982:38
- ⁹⁸Matthew 27:30, Mark 15:19, Luke 22:63, John 19:3
- ⁹⁹Meacham 1983:290, also Wilson 1979:43
- ¹⁰⁰Meacham 1983:286
- ¹⁰¹John 19:33-34
- ¹⁰²Edwards et. al. 1986:1461
- ¹⁰³*ibid.*
- ¹⁰⁴Barbet 1963:85-86
- ¹⁰⁵Lewis 1979:21
- ¹⁰⁶As quoted in Stevenson and Habermas 1990:109
- ¹⁰⁷Wilson 1979:34 mentions the "sensationalist claims of German writer Kurt Berna," made in the 1960s, and for a more modern example one might note *The Jesus Conspiracy* by Holger Kersten and Elmar Gruber (1992, Rockport, Mass.: Element).
- ¹⁰⁸Bucklin 1985:5
- ¹⁰⁹The postmortem issues of blood and the separation of serum at the wrists and the feet has been attributed to the removal of the crucifixion nails from the corpse. The postmortem flow visible on the left wrist is distinct from the flows on the forearm (which issued while the man was still alive on the cross). Stevenson and Habermas 1990:113.
- ¹¹⁰Presented most thoroughly in Heller and Adler 1981
- ¹¹¹Jackson 1988:3
- ¹¹²Barbet 1987a:4
- ¹¹³G. Lavoie et. al. 1986:4
- ¹¹⁴Bucklin 1982:36
- ¹¹⁵Pellicori and Evans 1981:43
- ¹¹⁶*ibid.*, p.41; Schwalbe and Rogers 1982:17
- ¹¹⁷Jackson 1988:4
- ¹¹⁸Jumper et. al. 1984:460, Heller and Adler 1981:91
- ¹¹⁹A detailed description of this technique is found in Whanger and Whanger 1985.
- ¹²⁰Whanger and Whanger 1991:3, Whanger and Whanger 1985:767
- ¹²¹Theories that the bulges were mere anomalies were rejected primarily because no such anomalies can be found elsewhere on the Shroud, and (possibly swollen) eyelids were deemed incapable of rendering the sharp fall-off at the edges of the bulges. Jackson et. al. 1977:89-90.
- ¹²²Whanger and Whanger 1991:3, Whanger and Whanger 1985:767, Stevenson and Habermas 1990:66
- ¹²³Whanger in Meacham 1983 Comments, p. 304.
- ¹²⁴John R. Cole in Meacham 1983 Comments, p. 296.
- ¹²⁵Whanger and Whanger 1985:770
- ¹²⁶Haralick 1983:34
- ¹²⁷Whanger and Whanger 1985:767, Whanger and Whanger 1991:4
- ¹²⁸Hachlili and Killebrew 1983
- ¹²⁹Reporting on the 1979 excavation of a large first century B.C.-A.D. Jewish cemetery in the hills overlooking Jericho, Rachel Hachlili wrote: "coins were also placed in the tombs. Two coins were found with coffin burials.... Two additional coins of Herod Agrippa I (41-44 A.D.) were found in a skull. *The coins originally must have been placed on the eyes of the deceased....*" (Hachlili 1979:34-35, emphasis added) Hachlili later recanted this statement in favor of the explanation that the coins were placed in the mouth as a vestige of the pagan custom of paying Charon, the mythological ferryman of the dead. This was the same explanation given in Greenhut 1992:35 for the presence of a bronze coin (dated 42/43 C.E.) in a woman's skull in the burial chamber of the Caiaphas family (that of the high priest who presided over the trial of Jesus). What makes this hypothesis rather untenable is the fact that a coin placed in the mouth of a corpse will not wind up in the skull; as the body decays, a coin so placed will fall among the vertebrae of the neck. Coins placed on the eyes, however, will fall through the orbital fissures into the skull (Alan Whanger in an unpublished letter to the editor of *BAR*, dated 15 September 1992, Mario Moroni in Berard 1991:280-281a). Such placement need not be explained by pagan mythology; both Jewish tradition and rabbinic writings of the Second Temple period prescribed the closing of the eyes of the deceased. In cases where rigor mortis would force the eyelids open, a small object placed on the lids would suffice to keep them shut. As the Pontius Pilate lepta were acceptable temple offerings (Mark 12:42, Luke 21:2,) there is no obvious reason why they could not have been used for this purpose.
- ¹³⁰Whanger and Whanger 1991:7
- ¹³¹*ibid.*
- ¹³²Haralick 1983:54, and in unpublished work.
- ¹³³Whanger and Whanger 1991:6
- ¹³⁴*ibid.* Of minor note is the indication in the phylactery on the left arm that the man on the Shroud was right-handed, a conclusion previously reached by Barbet in his studies of the shoulder abrasions and knee contusions (see section on pathology).
- ¹³⁵Wilson 1979:98-101; St. Augustine, *De Trinitate*, VIII, 4, 5, as quoted in Wilson 1979:101.
- ¹³⁶Wilson 1979:101-105
- ¹³⁷*ibid.*, p.105
- ¹³⁸Galey 1980:114-115 (plate 92)
- ¹³⁹Whanger and Whanger 1985:766-767
- ¹⁴⁰Whanger and Whanger 1991:10
- ¹⁴¹Stevenson and Habermas 1990:73
- ¹⁴²The reader is referred to Wilson's 1979 *The Turin Shroud*, which is the most thorough presentation available of the history of the Shroud prior to the fourteenth century.
- ¹⁴³Wilson 1979:136-172
- ¹⁴⁴*ibid.*, 172-191
- ¹⁴⁵Stevenson and Habermas 1981:17-18, Wilson 1978:30
- ¹⁴⁶Wilson 1979:155-172
- ¹⁴⁷There are exceptions. See previous note.
- ¹⁴⁸Wilson 1978:30

- ¹⁴⁹*ibid.* Wilson theorizes that somebody might have opened the frame the Mandylion was held in and seen the fold pattern without actually seeing the entire cloth.
- ¹⁵⁰Jackson 1984
- ¹⁵¹Maloney 1989:6 explains that "88 pollen grains were counted in approximately 2 square centimeters on a dorsal 'sidestrip' tape... 163 grains were counted on the same size area on a tape from the left arm, but an astounding ca. 300 grains were counted on a tape taken from near the face in a comparable size area."
- ¹⁵²The Veronica's veil legends seem to have arisen in 7th and 8th centuries, when the Mandylion/Shroud image had become widespread. Meacham 1983:287, Wilson 1979:106-112.
- ¹⁵³In his infamous memorandum, d'Arcis wrote: "since the holy Gospel made no mention of any such imprint... it was quite unlikely that the holy Evangelists would have omitted to record it..." (English translation by Rev. Herbert Thurston as originally published in *The Month*, CI, (A1903), pp.17-19.)
- ¹⁵⁴Daniel C. Scavone, "The History of the Turin Shroud to the 14th c." p.179, in Berard 1991:171-204; Pietro 1982; Wilson 1979:92-93.
- ¹⁵⁵Albert R. Dreisbach, Jr., "Did Peter See More Than an Empty Shroud?" in Berard 1991:345-357, and conversations with the author.
- ¹⁵⁶See Robinson 1977, Wuenschel 1945 and 1946. Some have incorrectly seen the account of Jesus' resurrection of Lazarus (John 11:38-44) as supporting the mummy-wrap theory. John, however, only writes that Lazarus' hands and feet were bound by *keiriai*, translated "twisted rushes." Had he been bound like a mummy, he would not have been able to proceed from the tomb without *first* being unbound. Evidence (a break in the blood flows at the wrist) has also been seen on the Shroud for a band tying the hands together.
- ¹⁵⁷Shabbath 23:5
- ¹⁵⁸The indications of a jaw-band are not the light strips bordering the face (see discussion of image formation) but the 3D reliefs which suggest (particularly on the left side) that the hair is draped over an object running round the head.
- ¹⁵⁹Pages 137-143 of Giulio Ricci's *The Holy Shroud* (1981, Italy: Giovanni Canestri).
- ¹⁶⁰"Duke Professor Unveils New Evidence on Shroud of Turin," Duke News Service press release dated 19 April 1984, pp.1-2.
- ¹⁶¹Luke 19:20, Acts 19:12
- ¹⁶²Matt. 27:57, Mark 15:42, Luke 23:54, John 19:42; Mark 16:1, Luke 24:1. See, for example, Wilson 1979:40-41; Robinson 1977:24-25; Wuenschel 1945:422-423.
- ¹⁶³B. Lavoie et. al. 1982:10-11. Lamm 1969:243-244 affirms this in the context of modern Jewish customs.
- ¹⁶⁴B. Lavoie et. al. 1982:12-15
- ¹⁶⁵Meacham 1986:15
- ¹⁶⁶Dmitri Kouznetsov in Wilson 1993/1994:5
- ¹⁶⁷Meacham 1986:20-23, Marino 1989:36
- ¹⁶⁸Schwalbe and Rogers 1982:41
- ¹⁶⁹Stevenson and Habermas 1990:65
- ¹⁷⁰Meacham 1986:21
- ¹⁷¹Emphasis his, as quoted in Marino 1989:36.
- ¹⁷²Stuckenrath 1965:281
- ¹⁷³Kouznetsov writing in Wilson 1993/94:3-4. As of this writing, he and Ivanov were still in the process of formally publishing their results.
- ¹⁷⁴Marie Claire Oosterwyck-Gastuche in Wilson 1993/1994:9
- ¹⁷⁵Meacham 1986:22
- ¹⁷⁶In Wilson 1993/1994:8
- ¹⁷⁷*ibid.*, p.12.
- ¹⁷⁸Gove 1987:193-194
- ¹⁷⁹Marino 1989:39,41-42; Stevenson and Habermas 1990:57
- ¹⁸⁰As quoted in Marino 1989:42
- ¹⁸¹Marino 1989:42
- ¹⁸²Page 2 of Clark, Kenneth R. 1988. Carbon Tests Prove Shroud is Not Burial Cloth of Jesus. *Chicago Tribune*, 14 October, pp.1-2.
- ¹⁸³Miller and Pellicori 1981, Marino 1989:36-37
- ¹⁸⁴Miller and Pellicori 1981:85, Schwalbe and Rogers 1982:42
- ¹⁸⁵Stevenson and Habermas 1990:58
- ¹⁸⁶Meacham 1986:16
- ¹⁸⁷Wilson 1988:7 reprints the text of the certificate handed over to the three laboratories along with the specimens. The certificate states quite clearly the sources and dates of the control cloths used, and since the weaves of the two control samples were quite distinct from each other and from the Shroud, and the origin of each could be easily identified, any claims to "blindness" in the testing are rather absurd. See also Marino 1989:38, Stevenson and Habermas 1990:50-51,57.
- ¹⁸⁸Marino 1989:43, Wilson 1988:6
- ¹⁸⁹Marino 1989:38, Stevenson and Habermas 1990:56
- ¹⁹⁰Oosterwyck-Gastuche in Wilson 1988:11-12, Remi Van Haelst in privately printed manuscript "Radiocarbon Dating the Shroud of Turin."
- ¹⁹¹As quoted in Marino 1989:40
- ¹⁹²Meacham 1986:18
- ¹⁹³Damon et. al. 1989:614
- ¹⁹⁴Tite 1986:571
- ¹⁹⁵Marino 1989:38-39
- ¹⁹⁶Wolff, with others, as quoted in Meacham 1986:24, emphasis added.
- ¹⁹⁷As quoted in Stevenson and Habermas 1990:56, emphasis theirs.
- ¹⁹⁸Schwalbe and Rogers 1982:45
- ¹⁹⁹"Preliminary results of the 1978 Shroud of Turin investigation," Brochure from the Visual Data Exhibition at the University of Connecticut (11 October - 9 December 1981,) pp.1-2.
- ²⁰⁰See in particular Schwalbe and Rogers 1982 and Jackson, Jumper, and Ercoline 1984.
- ²⁰¹Schwalbe and Rogers 1982:9-24; Heller and Adler 1981; Jumper et. al. 1984; Jackson, Jumper, and Ercoline 1984:2251-2253.
- ²⁰²See Stevenson and Habermas 1990:123.
- ²⁰³Gilbert and Gilbert 1980:1935
- ²⁰⁴Miller and Pellicori 1981:84
- ²⁰⁵Schwalbe and Rogers 1982:25-27
- ²⁰⁶There are certain lateral 2D distortions on the Shroud image attributable to cloth drape. (See Ercoline et. al. 1982.)
- ²⁰⁷Jackson, Jumper, and Ercoline 1984:2253,2260,2263
- ²⁰⁸Jackson et. al. 1977:82-83, Pellicori and Evans 1981:84-85
- ²⁰⁹Schwalbe and Rogers 1982:34-35
- ²¹⁰This includes the Pellicori-German latent image hypotheses, which assert that the image was formed over time (possibly as the cloth began to sag) as the cellulose in the cloth reacted with sensitizing materials transferred by contact with a corpse. Whatever chemical and density-shading merits these theories have, they still introduce the distortion and saturation problems associated with direct contact. Jackson, Jumper, and Ercoline 1984:2263,2265; Schwalbe and Rogers 1982:28-29,33-36; Stevenson and Habermas 1990:125-128.
- ²¹¹Schwalbe and Rogers 1982:32-33
- ²¹²Schwalbe and Rogers 1982:35; Jackson, Jumper, and Ercoline 1984:2264; Stevenson and Habermas 1990:124-125
- ²¹³Jackson, Jumper, and Ercoline 1984:2264
- ²¹⁴G. R. Lavoie et. al. 1983
- ²¹⁵Jackson 1988:9
- ²¹⁶Jackson 1989
- ²¹⁷*ibid.*, p.15
- ²¹⁸Jackson 1988:9
- ²¹⁹Jackson 1990:5
- ²²⁰*ibid.*, p.10
- ²²¹Carter 1984:433
- ²²²*ibid.*, pp.444-446
- ²²³Jumper et. al. 1984:451-453, Schwalbe and Rogers 1982:24
- ²²⁴Jumper et. al. 1984:467, Heller and Adler 1981:97
- ²²⁵Carter 1984:439-440
- ²²⁶*ibid.*, pp.431-433.
- ²²⁷As quoted in press release dated 21 March 1994.
- ²²⁸"An update on 'Why Science Cannot Cope with the Shroud of Turin'", text of a lecture presented at the University of Toronto, 2 March 1993, pp.9-10.
- ²²⁹*ibid.*, p.12.
- ²³⁰Whanger and Whanger 1991:4
- ²³¹"Duke Professor Unveils New Evidence on Shroud of Turin," Duke University News Release dated 19 April 1984, p.3.
- ²³²"An update...", pp.12, 16-20.
- ²³³As quoted on page 35 of Zurer, Pamela S. 1983. Archaeological Chemistry. *Chemistry and Engineering News*, 21 February, pp. 26-44.
- ²³⁴This is precisely why no first-century Jewish burial shrouds from the Jerusalem area are known to exist; while there are thousands of Egyptian shrouds in existence due to the drier climate and consequent preservation of bodies, the relative humidity of the Jerusalem area leads to more rapid decomposition.
- ²³⁵Bucklin 1985:6.
- ²³⁶Cadaveric spasm can slow the loss of rigor mortis, and environmental conditions affect the rate of decomposition, so it is difficult to give an absolute time for the removal of the body. Meacham 1983:285, Stevenson and Habermas 1990:138.
- ²³⁷As quoted on page 65 of Murphy, Cullen. 1981. Shreds of evidence. *Harper's*, November, pp. 42-65.
- ²³⁸Text of address reprinted in *Shroud Spectrum International* (December 1993) and as a special monograph for the British Society for the Turin Shroud (March/April 1994).
- ²³⁹A detailed proposal for future studies was drafted by the Board of Directors of STURP in 1984 and is printed as D' Muhala, Thomas, J. Jackson, W. Ercoline, A. Adler, R. Dichtl, R. Dinegar, E. Jumper. 1984. A Sci-

- entific Proposal for Studying the Shroud of Turin. *Shroud Spectrum International*, No.13(December):9-13. The pre-carbon dating proposal enumerates 83 specific questions to be investigated, almost all of which must remain unaddressed until further direct studies of the cloth are permitted. For a more recent outline for further studies, see Schwalbe, L. A. 1990. Scientific Issues and Shroud Research in the 1990s. *Shroud Spectrum International*, No.35/36(June/Sept.):2-12.
- ²⁴⁰Meacham 1983:294
- ²⁴¹The Roman emperor Constantine banned crucifixion in 315 C.E., approximately five hundred years after it had first been practiced by the Romans.
- ²⁴²While team members are far from agreeing on how the image actually got on the cloth, the artistry hypothesis is uniformly rejected.
- ²⁴³It is intriguing that of all who believe the Shroud a work of art, McCrone has also changed his mind the most frequently: he has claimed at various times that the Shroud image is a pigment enhancement of an actual image; the result of a synthetic iron oxide; due to iron earths and vermillion.
- ²⁴⁴Letter to the editor of *The Microscope*, 30:348.
- ²⁴⁵Nickell, Joe. 1994. Pollens on the 'Shroud': A Study in Deception. *The Skeptical Inquirer* 18(3): 379-385.
- ²⁴⁶In Meacham 1983 "Comments," p.298.
- ²⁴⁷See Nickell's book *Inquest on the Shroud of Turin* (1983, Buffalo: Prometheus Press).
- ²⁴⁸In Meacham 1983 "Comments," p.299.
- ²⁴⁹Craig and Bresee 1994
- ²⁵⁰Associated Press News Release dated 10 February 1994, written by Duncan Mansfield.
- ²⁵¹Craig and Bresee 1994:67
- ²⁵²*ibid.*
- ²⁵³See, for example, page 169 of John Cornwell's *The Hiding Places of God*. (1991, New York: Warner Books). The author has also heard someone argue this on the Internet, under the posting "Shroud of Turin - Smoking Gun - Jewicide."
- ²⁵⁴Robinson 1977:23
- ²⁵⁵As quoted in Wilson 1988:5.
- ²⁵⁶Giulio Ricci in Marino 1989:44
- ²⁵⁷Stevenson and Habermas 1990:69-71 mention the work of Fossati, Crispino, Morgan, Currer-Briggs, and others, much of which has been printed or reprinted in *Shroud Spectrum International*.
- ²⁵⁸This is allowing for a several decade circulation of Pontius Pilate lepta and Tiberian amulets. It is a major stretch to assert that somebody several centuries later would have procured two lepta and an amulet from the reign of a previous emperor, especially when the presence of these items on the body of a crucified Jew was hardly typical.
- ²⁵⁹Meacham 1983:291
- ²⁶⁰As quoted in Stevenson and Habermas 1990:179
- ²⁶¹Meacham 1983:292
- ²⁶²*ibid.*, p.293
- ²⁶³Morgan, Rex. 1986. *The Holy Shroud and the Earliest Paintings of Christ*. Hong Kong: The Runciman Press.
- ²⁶⁴Meacham 1982:309
- ²⁶⁵As quoted in Wilson 1979:33-34 and Van Cauwenberghe 1992:17.
- ²⁶⁶Vignon's vaporograph theory was the prototypical diffusion theory.
- ²⁶⁷Jumper et. al. 1984:474
- ²⁶⁸Schwalbe and Rogers 1982:45
- ²⁶⁹Stevenson and Habermas 1981:14

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Connie: Well, I would certainly agree with all that. When I first met the Lord, I met Him through reading the Word. And I knew at that time that He knew everything about me, and yet loved me anyhow. I told Him then that if He would accept me as I was that I would try to live my life His way. To me, Christianity really means that - it means having a personal relationship with the Lord Jesus Christ, and really leaving my life in His hands. That relationship with Him is nurtured through reading the Bible, through prayer, on a daily basis. I think one of

the real strengths of my relationship with Bill and of our marriage is that we read the Word and we pray together twice a day. We've done that for over eighteen years now, and it makes a difference. And so, what being a Christian means to me personally is that I look to Christ for guidance, I look to Him for direction, I look to Him for strength, in really every aspect of my life. Whether I am in a worship service or doing scientific research, Jesus is the central fact, the central feature of my life.

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